

LLMs Are Not the Answer

Relevant AI for Business Transformation

Christopher Simpkins

<https://chris.simpkins.phd/>

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Businesses “Adopting” AI



Distracted boyfriend meme; chasing ChatGPT instead of the world of relevant and useful AI

Get on the hype train before you're left behind!



Oofda. The hype train wasn't so great.



95% of Generative AI Projects in Businesses Fail



The GenAI Divide

STATE OF AI IN BUSINESS 2025

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¹https://www.artificialintelligence-news.com/wp-content/uploads/2025/08/ai_report_2025.pdf

Whatever. I got my bag!



What is Artificial Intelligence?

Intelligence is the ability to do:^{2, 3, 4, 5, 6}

1. **Problem solving**: transforming a given state into a goal state.
2. **Inference**: deriving new knowledge from existing knowledge.
3. **Reasoning**: answering questions or justifying answers.
4. **Decision making**: choosing actions that maximize expected utility.
5. **Learning**: using experience (e.g., data) to get better at a task, such as those listed above.

A human-made (*artificial*) system that does any of the above is an AI system.

²<https://www.apa.org/topics/intelligence>

³<https://www1.udel.edu/educ/gottfredson/reprints/1997mainstream.pdf>

⁴<https://www.richardhaier.com/books/the-neuroscience-of-intelligence>

⁵<https://mitpress.mit.edu/9780262537537/the-sciences-of-the-artificial/>

⁶<https://aima.cs.berkeley.edu/>

Machine Learning as a Technique

Learning distinguishes intelligence, but it's not a type of task. It's a technique for designing systems that accomplish tasks.

1. ML problem solving: randomized algorithms for optimization, e.g., genetic algorithms, simulated annealing, information-theoretic generate-test
2. ML inference: supervised ML, e.g., linear and logistic regression, SVMs, ANNs, decision trees; unsupervised ML, e.g., clustering, PCA
3. ML decision making: reinforcement learning

Use machine learning when it's hard to define an algorithm or write a set of rules, but you have experience (data) that contains patterns that a machine learning algorithm can discover.

95% of deployed AI systems are "classical AI," not machine learning systems. – AAAI 20-Year Roadmap, AAAI-19 Conference

What about the AI hype train?!



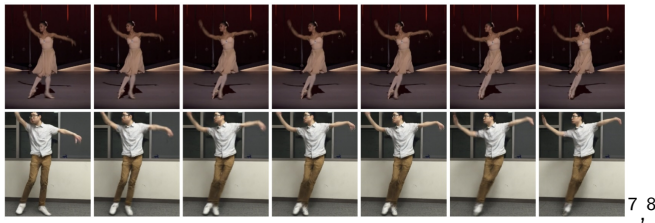
What is “generative AI?”

Two types of supervised statistical machine learning:

- ▶ Discriminative, $P(\mathbf{y} | \mathbf{X})$: learn a function that discriminates between classes
- ▶ Generative, $P(\mathbf{X}, \mathbf{y})$: learn a joint probability distribution over data and labels
 - ▶ Enables generative models to both discriminate and *generate samples*

Modern GenAI based on deep learning (neural nets with > 1 hidden layer, typically $\gg 1$)

- ▶ Gained attention with Ian Goodfellow’s generative adversarial networks (GANs) in 2014, now auto-regressive transformer models all the rage, e.g., large language models (LLMs) like ChatGPT

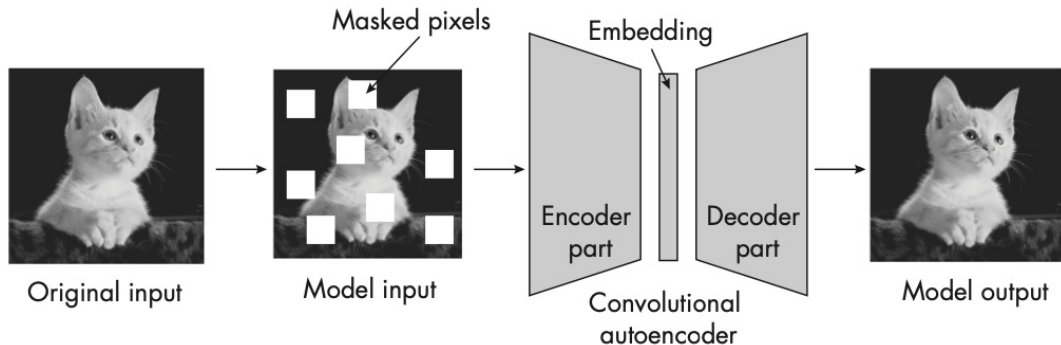


⁷<https://www.youtube.com/watch?v=PCBTZh41Ris&t=173s>

⁸<https://arxiv.org/pdf/1808.07371.pdf>

You've been using GenAI for years.

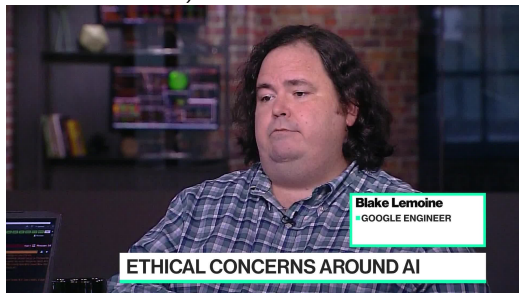
E.g., automatically fixing images.



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You've even been using LLMs for years.

- ▶ Modern LLMs based on transformer architecture developed for machine translation.
 - ▶ $\{en, fr, de, \dots\} \rightarrow \{en, fr, de, \dots\}$
- ▶ Google had a chatbot for years but didn't release it publicly.
 - ▶ Remember when Blake Lemoine grabbed his 15 minutes of fame in 2021 (long before ChatGPT was released)?



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¹⁰ <https://www.youtube.com/watch?v=kgCUn4fQTsc>

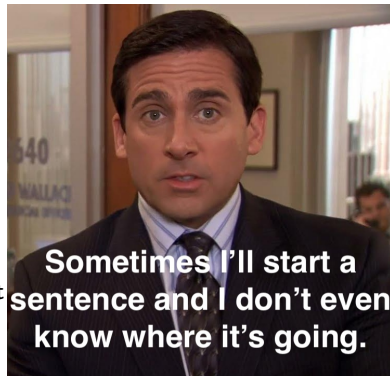
Large Language Models (LLMs) are Stochastic Parrots

An LLM predicts next “words” given previous words, including its own output – they are auto-regressive.

$$\prod_{t=1}^T p(word_t | word_{1:t-1})$$

- ▶ LLMs are sub-symbolic models of text sequences with billions of parameters (floating-point numbers) “trained” on terabytes of data.
 - ▶ **Training is a technical term meaning iteratively adjusting the parameters of a predictive model. It has nothing to do with what we call training for human skills.**
 - ▶ Training costs millions of dollars and takes months to years.
 - ▶ Models “frozen in time.”
- ▶ Generated “answers” nothing more than statistically plausible sequences of text based on patterns in training data and prompt.
 - ▶ LLMs don't give correct or incorrect answers.
 - ▶ Every response is just a probabilistic event.

"Hallucination" is a marketing term – a creative lie.



“But they will fix hallucinations!”

No, they won't. It's like saying you'll keep adjusting the weighting of dice until they reliably roll what you “want.”



Dr. CS's take: generative AI can be useful. The technology behind LLMs is useful. But LLM-based chatbots like ChatGPT and LLM-based programming tools are not only useless, but harmful.

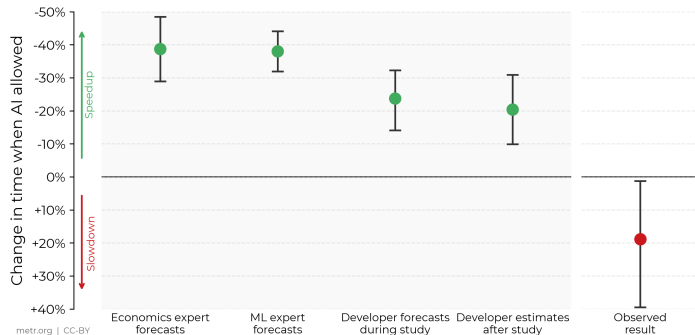
“But I’m more productive with LLMs!”

No, you are not!

Against Expert Forecasts and Developer Self-Reports, Early-2025 AI Slows Down Experienced Open-Source Developers



In this RCT, 16 developers with moderate AI experience complete 246 tasks in large and complex projects on which they have an average of 5 years of prior experience.



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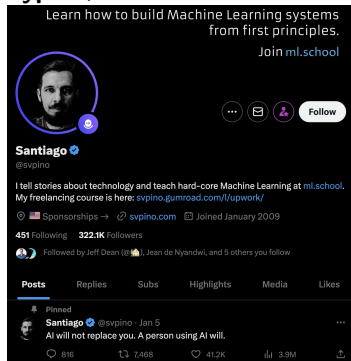
- Even highly skilled people *just want to believe* and can't be convinced otherwise.
We're fighting not just sociopath grifters in Silicon Valley, but human nature.

¹¹<https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/>

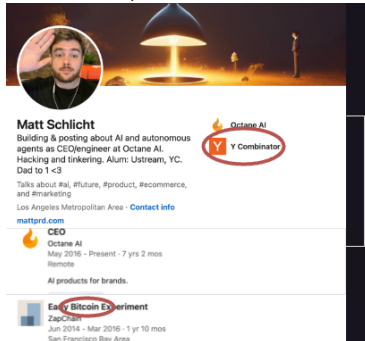
The loudest mouths will hail the new found way.¹²

Unfortunately, the public discourse is filled with noise from various grifters:

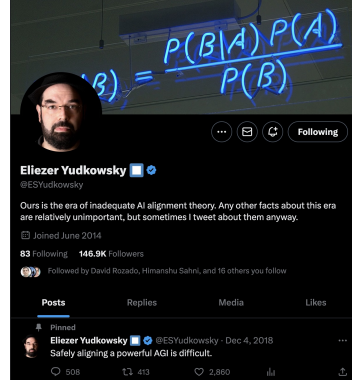
hypers,



charlatans, and



doomers.



- ▶ This noise renders much of the public who don't know technical details of AI deaf to reason.
- ▶ Anyone who says AGI is around the corner knows nothing about AI or is selling something.

¹²[https://en.wikipedia.org/wiki/King_for_a_Day_\(XTC_song\)](https://en.wikipedia.org/wiki/King_for_a_Day_(XTC_song))

Experts can by hypers too.



The screenshot shows a tweet thread on a dark background. The top tweet is from Jaana Dogan (@rakyll), a verified user, with a blue checkmark. Her text reads: "To cut through the noise on this topic, it's helpful to provide more more context: - We have built several versions of this system last year. - There are tradeoffs and there hasn't been a clear winner. - When prompted with the best ideas that survived, coding agents are able to go very far and generate a good decent toy version in an hour or so." Below this is a retweet from Rohan Paul (@rohanpaul_ai), also verified, dated Jan 3. His text says: "Dario Amodei was so right about AI taking over coding. This is from a Principal Engineer at Google. Claude Code matched 1 year of team output in 1 hour..." with a "Show more" link. Below that is another tweet from Jaana Dogan (@rakyll), dated Jan 3, 2026, at 5:27 AM, with 3.8M views. Her text reads: "I'm not joking and this isn't funny. We have been trying to build distributed agent orchestrators at Google since last year. There are various options, not everyone is aligned... I gave Claude Code a description of the problem, it generated what we built last year in an hour." The bottom of the screenshot shows the tweet's timestamp as 10:45 PM on Jan 3, 2026, with 280K views.

Jaana Dogan ヤナ ドガン ✓
@rakyll

To cut through the noise on this topic, it's helpful to provide more more context:

- We have built several versions of this system last year. - There are tradeoffs and there hasn't been a clear winner.
- When prompted with the best ideas that survived, coding agents are able to go very far and generate a good decent toy version in an hour or so.

Rohan Paul ✓ @rohanpaul_ai · Jan 3

Dario Amodei was so right about AI taking over coding.

This is from a Principal Engineer at Google.

Claude Code matched 1 year of team output in 1 hour...
[Show more](#)

Jaana Dogan ヤナ ドガン ✓
@rakyll

I'm not joking and this isn't funny. We have been trying to build distributed agent orchestrators at Google since last year. There are various options, not everyone is aligned... I gave Claude Code a description of the problem, it generated what we built last year in an hour.

5:27 AM · Jan 3, 2026 · **3.8M** Views

10:45 PM · Jan 3, 2026 · **280K** Views

People post nonsense like this constantly.

- ▶ Jaana Dogan is a principal engineer at Google.
 - ▶ She knows better!
- ▶ How is a junior engineer supposed to filter the noise?

[https:](https://x.com/rakyll/status/2007659740126761033)

[//x.com/rakyll/status/2007659740126761033](https://x.com/rakyll/status/2007659740126761033)

<https://x.com/rakyll/status/2007659740126761033>

Concluding Word on Generative AI

The reality is that ChatGPT is an impressive toy. Watch ~10 minutes from the 47-minute mark in this talk from Yann LeCun: <https://www.youtube.com/watch?v=pd0JmT6rYcl>

“In a few years, nobody in their right mind will use auto-regressive LLMs [like GPT].” – Yann LeCun, Turing Laureate and Deep Learning Pioneer

- ▶ AI is powerful and useful. It has been for decades.
- ▶ The current LLM craze dominates the popular media, but is a small part of AI.
- ▶ The key to getting value from AI is to *drive AI adoption from business problems*, not the other way around.
 - ▶ Choose the right AI solution for a given problem, or no AI at all when that's the right call.

AI Engineering for Business

AI systems (like other systems) must solve business problems.

Foundation of all business problems: maximize profit.

- ▶ Profit = Revenue - Cost

Business (AI) systems prove value by:

- ▶ increasing revenue, or
- ▶ decreasing cost.

AI systems must provide clear business value, just like any other system.

Technology Interlude: Vector Space Document Embedding (Intuition)

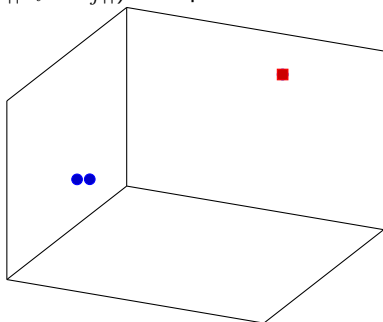
Imagine term by document matrix containing normalized counts of words in documents.

	d_1	d_2	$\dots$	d_n
shoulder	0.20	0.20	$\dots$	0.01
arm	0.19	0.00	$\dots$	0.01
manipulation	0.00	0.21	$\dots$	0.01
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
pancreas	0.19	0.20	$\dots$	0.02

Then each document is a vector ...

$$\begin{bmatrix} 0.20 \\ 0.19 \\ 0.00 \\ \vdots \\ 0.19 \end{bmatrix} \begin{bmatrix} 0.20 \\ 0.00 \\ 0.21 \\ \vdots \\ 0.20 \end{bmatrix} \dots \begin{bmatrix} 0.01 \\ 0.01 \\ 0.01 \\ \vdots \\ 0.02 \end{bmatrix}$$

... and distance between vectors (e.g., $||\vec{d}_i - \vec{d}_j||$) corresponds to semantic similarity.

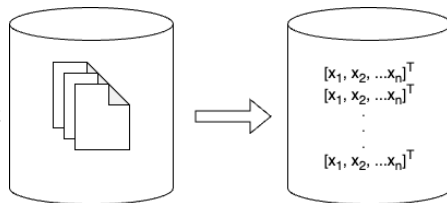


Documents don't need same terms. "shoulder" and "arm" will co-occur with many same words. This idea is known as *latent semantic indexing*. Note: this is highly simplified for illustration.

Document Matching

Technology:

- ▶ Scan documents, create document vectors.
- ▶ Store document vectors in vector database.
- ▶ Vector database provides “similarity search” based on document vector distances.



Applications:

- ▶ General search (many off-the shelf options, e.g., Lucene)
- ▶ Document association (need a better term. You have documents in separate repositories that are “about” the same thing and you want a system to automatically retrieve related documents.)

Dump

- ▶ Predictive maintenance
- ▶ Customer segmentation via clustering
- ▶ Collaborative filtering
- ▶ A/B testing via bandits
- ▶ Supply chain optimization via RL